

Data Analysis with Python

Full tutorial for beginners

IMPORTANT: We're in the process of adapting the projects into DataWars.io for your simplicity.

(stay tuned 👍)

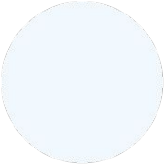
About this tutorial

1. What is Data Analysis
2. [Real example Data Analysis with Python](#)
3. [How to use Jupyter Notebooks](#)
4. [Intro to NumPy \(exercises included\)](#)
5. [Intro to Pandas \(exercises included\)](#)
6. [Data Cleaning](#)
7. [Reading Data SQL, CSVs, APIs, etc](#)
8. [Python in Under 10 Minutes](#)





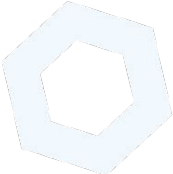
● **What is Data Analysis?**

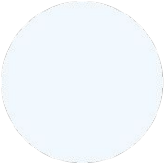


What is Data Analysis

> A process of inspecting, cleansing, transforming and modeling data with the goal of discovering useful information, informing conclusion and supporting decision-making.

[Definition by Wikipedia.](#)

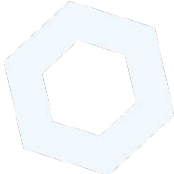


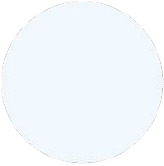


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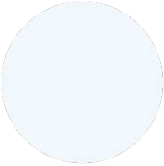


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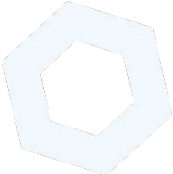


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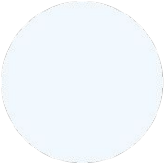


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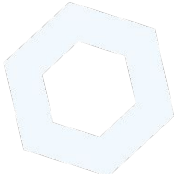
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Three small teal spheres are positioned on the slide: one to the left of the title, one in the upper right corner, and one in the lower right corner.

Data Analysis Tools

Auto-managed closed tools



Programming Languages



Auto-managed closed tools

👎 Closed Source 🙈

👎 Expensive 💰

👎 Limited 😞

👍 Easy to learn 🙈

Programming Languages

👍 Open Source 🤩

👍 Free (or very cheap) 😄💰

👎 Extremely Powerful 💪

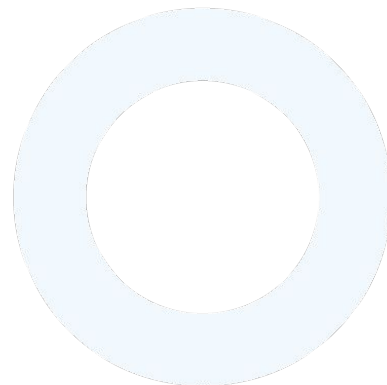
👎 Steep learning curve 🙈

Why Python for Data Analysis?

Why Python for Data Analysis?

Why would we choose Python over R or Julia?

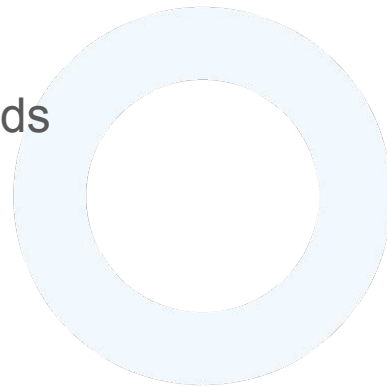
- 👍 very simple and intuitive to learn
- 👍 “correct” language
- 👍 powerful libraries (not just for Data Analysis)
- 👍 free and open source
- 👍 amazing community, docs and conferences



When to choose R?

Python, sadly, is not always the answer

- When R Studio is needed
- When dealing with advanced statistical methods
- When extreme performance is needed



The Data Analysis Process



Data Extraction

- SQL
- Scrapping
- File Formats
 - CSV
 - JSON
 - XML
- Consulting APIs
- Buying Data
- Distributed Databases

Data Cleaning

- Missing values and empty data
- Data imputation
- Incorrect types
- Incorrect or invalid values
- Outliers and non relevant data
- Statistical sanitization

Data Wrangling

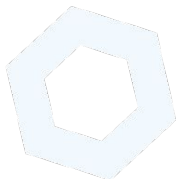
- Hierarchical Data
- Handling categorical data
- Reshaping and transforming structures
- Indexing data for quick access
- Merging, combining and joining data

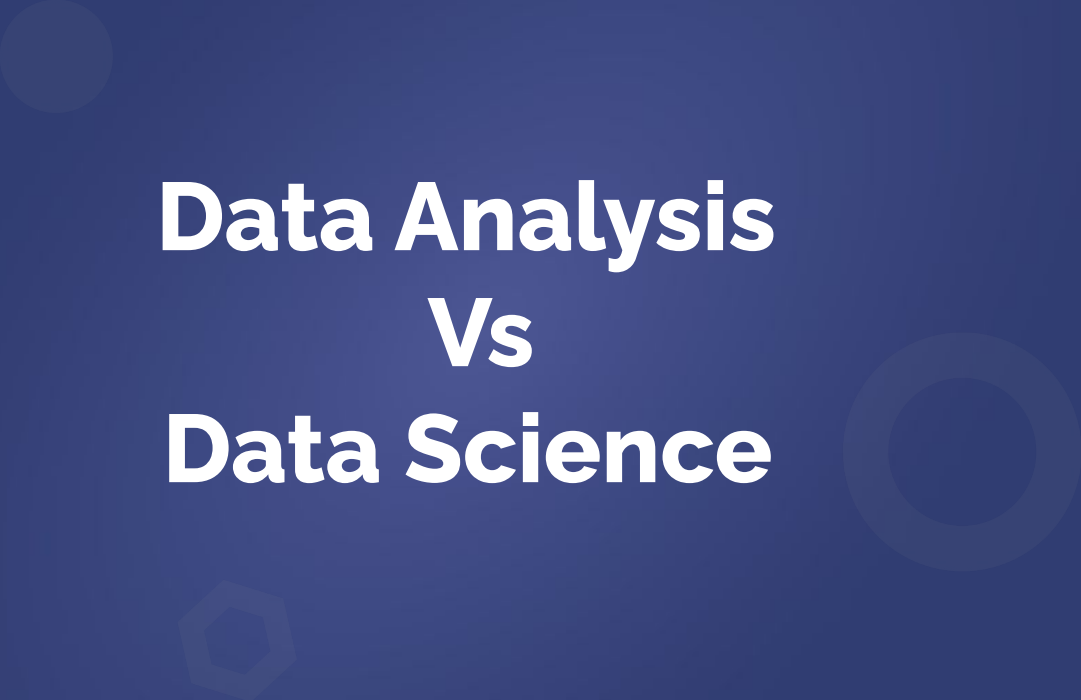
Analysis

- Exploration
- Building statistical models
- Visualization and representations
- Correlation vs Causation analysis
- Hypothesis testing
- Statistical analysis
- Reporting

Action

- Building Machine Learning Models
- Feature Engineering
- Moving ML into production
- Building ETL pipelines
- Live dashboard and reporting
- Decision making and real-life tests

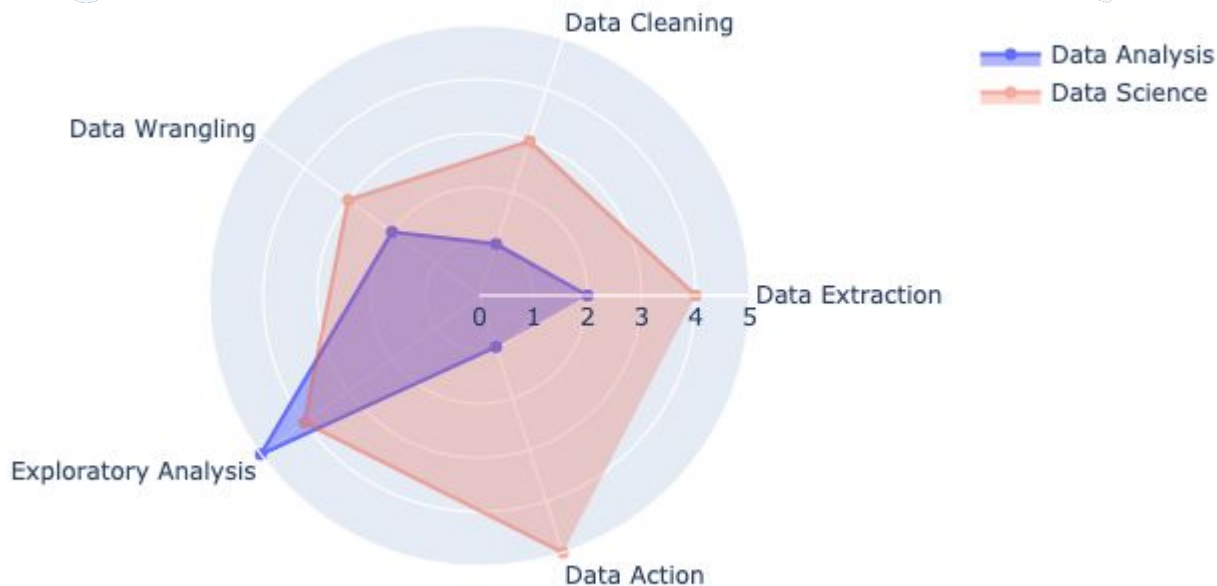




Data Analysis Vs Data Science

DATA ANALYSIS VS DATA SCIENCE

The traditional view



Python & PyData Ecosystem

PYTHON ECOSYSTEM:

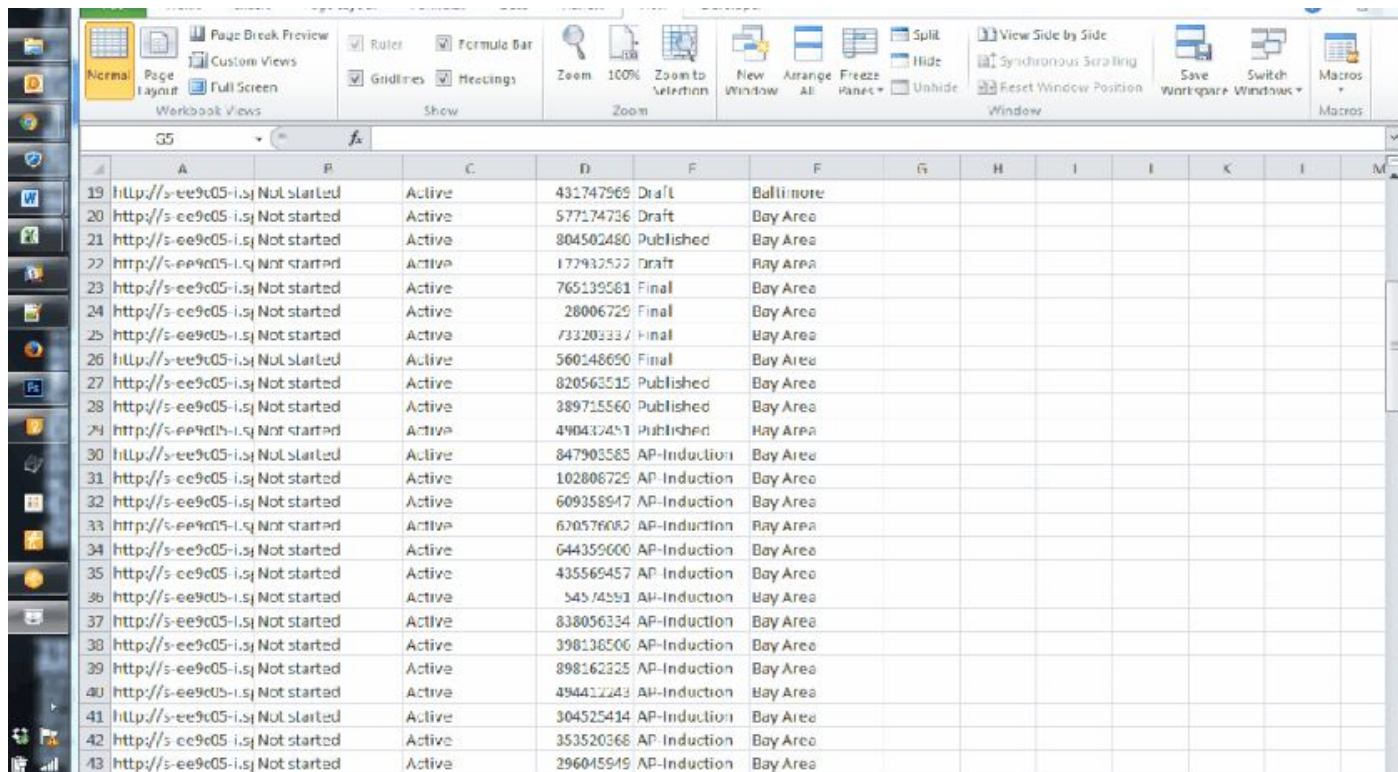
The libraries we use...

- [pandas](#): The cornerstone of our Data Analysis job with Python
- [matplotlib](#): The foundational library for visualizations. Other libraries we'll use will be built on top of matplotlib.
- [numpy](#): The numeric library that serves as the foundation of all calculations in Python.
- [seaborn](#): A statistical visualization tool built on top of matplotlib.
- [statsmodels](#): A library with many advanced statistical functions.
- [scipy](#): Advanced scientific computing, including functions for optimization, linear algebra, image processing and much more.
- [scikit-learn](#): The most popular machine learning library for Python (not deep learning)

How Python Data Analysts Think

EXCEL, TABLEAU, ETC.

They're all visual tools...



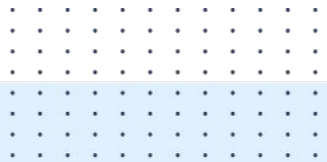
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Thinking like a Python Data Analyst



**And finally,
why Python?**





>20%

Salary increase for a Data Analyst
that knows Python and SQL.



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